

Lab Test II: Smart Microwave Oven Microcontroller Design

(FSM + Interrupts + Timing + User Interaction)

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1. Objective

The objective of this assignment is to design and simulate a **microcontroller-based control system** for a Smart Microwave Oven.

Students will learn:

- FSM-based control design
- Interrupt handling
- Timer-based execution
- Embedded system simulation

2. System Description

You are required to design a controller for a microwave oven.

Inputs

- START
- STOP
- DOOR_CLOSED (1 = closed, 0 = open)
- FOOD_PRESENT (1 = present)
- TEMP_SENSOR (1 = overheating)

Outputs

- HEATER (ON/OFF)
- TURNTABLE (ON/OFF)
- LIGHT (ON/OFF)
- BUZZER (ON/OFF)
- DISPLAY (Mode + Time)

3. Operating Modes

Mode	Description	Time (cycles)
DEFROST	Low power	3
HEAT	Medium power	5
GRILL	High power	7

4. FSM States

IDLE → CHECK → RUN → PAUSE → COMPLETE → BUZZ → IDLE
 ↓
 ERROR

5. State Behavior

State	HEATER	TURNTABLE	LIGHT	BUZZER
IDLE	OFF	OFF	OFF	OFF
CHECK	OFF	OFF	ON	OFF
RUN	ON	ON	ON	OFF
PAUSE	OFF	OFF	ON	OFF
COMPLETE	OFF	OFF	ON	OFF
BUZZ	OFF	OFF	OFF	ON
ERROR	OFF	OFF	ON	ON

6. Functional Rules

Start Condition

System starts only if:

- START = 1
- DOOR_CLOSED = 1
- FOOD_PRESENT = 1

Pause Condition

- STOP pressed → PAUSE
- Resume using START

Completion

- When timer reaches zero → COMPLETE → BUZZ

7. Interrupt Handling

Interrupt Priority

TEMP_SENSOR > DOOR_OPEN > STOP

Interrupt Behavior

- TEMP_SENSOR = 1 → ERROR state immediately
- DOOR_CLOSED = 0 → PAUSE immediately

8. Timing Rules

- Each state = 1 cycle
- RUN duration depends on selected mode
- Timer decreases every cycle in RUN

9. Display Requirement

At each cycle:

DISPLAY = MODE - Remaining Time

Example:

HEAT - 3

10. Execution Model

At each cycle:

1. Read inputs
2. Check interrupts
3. Execute state
4. Update outputs
5. Update timer
6. Print log

11. Output Format (Mandatory)

Cycle | State | Inputs | Outputs | Display

Example:

Cycle 3 | RUN | DOOR=1 | HEATER=ON | HEAT-3

12. Sample Test Cases

Test Case 1: Normal Operation

MODE = HEAT

Cycle 1: START=1, DOOR=1, FOOD=1

Expected Output (Key Steps)

Cycle 2 | RUN | HEATER=ON | HEAT-5

Cycle 3 | RUN | HEAT-4

Cycle 7 | COMPLETE

Cycle 8 | BUZZ

Test Case 2: Door Open During Run

Cycle 4: DOOR=0

Expected Output

Cycle 4 | PAUSE | HEATER=OFF

Test Case 3: Resume

Cycle 5: START=1

Expected Output

Resume from remaining time

Test Case 4: Overheat

Cycle 3: TEMP_SENSOR=1

Expected Output

Cycle 3 | ERROR | BUZZER=ON

Test Case 5: Invalid Start

START=1, DOOR=0

Expected Output

System remains in IDLE

13. Tasks

- Draw microcontroller block diagram
- Draw FSM diagram
- Write C/C++ simulation
- Show outputs for all test cases

14. Important Notes

- Outputs depend only on current state
- Inputs may change during execution
- Exact cycle numbers may vary slightly, but logic must match